Deduce "what it is" from the curve form (Fsou classif, D2)

$\boxed{\text{ELLIPSE}}$ $C: \underbrace{73x^2+72xy+52y^2}_{Q}-10x+55y+25=0$

– not clear that C is an ellipse –

→ symm. matrix: $Q: \begin{bmatrix} 73 & \frac{72}{2} \\ \frac{72}{2} & 52 \end{bmatrix} = \begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix}$
$(A=A^T)$

→ $C: [x \ y] \underbrace{\begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix}}_{Q} \begin{bmatrix} x \\ y \end{bmatrix} + \cancel{\begin{bmatrix} -10 \\ 55 \end{bmatrix}} \underbrace{[-10 \ \ 55]}_{b} \begin{bmatrix} x \\ y \end{bmatrix} + 25 = 0$

→ eigenvalues of Q:

$\begin{vmatrix} 73-\lambda & 36 \\ 36 & 52-\lambda \end{vmatrix} = (73-\lambda)(52-\lambda)-36^2=0 \Rightarrow \dots \Rightarrow \lambda_1=100$
$\Rightarrow \lambda_2 = 25$

→ eigenvectors:

$(Q-100I_2)\overset{v_1}{} = 0 \Rightarrow \left(\begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix} - \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}\right) \vec{v} = 0$

$\Rightarrow \begin{bmatrix} -27 & 36 \\ 36 & -48 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{cases} -27x+36y=0 \ /:9 \\ 36x-48y=0 \ /:6 \end{cases} \Rightarrow \begin{cases} -3x+4y=0 \\ 6x-8y=0 \end{cases}$

$\Rightarrow \begin{cases} -3x+4y=0 \\ 3x-4y=0 \end{cases} \Rightarrow 3x=4y \Rightarrow x=\frac{4y}{3} \Rightarrow \begin{bmatrix} \frac{4y}{3} \\ y \end{bmatrix}$

$\Rightarrow y=\frac{3}{4}x$

choose $x=4 \Rightarrow y=3 \Rightarrow (4,3)=v_1$.

$(Q-25I_2)v_2=0 \Rightarrow \dots \quad v_2=(3,-4)$.

$\Rightarrow v_1, v_2$ form an ORTHOGONAL BASIS $\Rightarrow \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$ orthogonal matrix
$(v_1 \perp v_2)$ det $= -16-9=-25$.

→ we want ORTHONORMAL BASIS, matrix because it is a ROTATION MATRIX

$\Rightarrow$ det $=1 \Rightarrow$ preserves orientation, lengths

$B' = (e_1'(\frac{4}{5}, \frac{3}{5}), e_2'(\frac{3}{5}, -\frac{4}{5}))$ orthonormal basis

$M_{\underset{\sim}{BB'}} = \frac{1}{5}\begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$ → orthogonal matrix $(M^{-1}=M^T)$, det $=-1 \Rightarrow$ indirect isometry.

"Normalization" = divide each entry of the matrix of eigenvectors $\begin{bmatrix} v_1 & v_2 \\ \vdots & \vdots \end{bmatrix}$ by the length of each eigenvector

$\|v_1\| = \sqrt{4^2+3^2} = \sqrt{25} = 5$ → divide column 1 by 5.

$\|v_2\| = \sqrt{3^2+(-4)^2} = 5$ → divide column 2 by 5.

$\det M = -1 \neq 0 \Rightarrow$ invertible matrix $\Rightarrow$ IS TRANSFORMATION.

↳ INDIRECT ISOMETRY

→ make it DIRECT:

$$\underset{\sim}{M_{BB'}} = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \in SO(2)$$

$$\mathcal{C}: [x' \ y']\, M_{BB'}^{T} \cdot Q \cdot M_{BB'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + b \cdot M_{BB'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + 25 = 0.$$

$$\mathcal{C}: [x' \ y'] \cdot \frac{1}{25}\begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix} \cdot \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \cancel{\begin{bmatrix} x' \\ y' \end{bmatrix}} + b \cdot \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix} + 25 = 0$$

$$\mathcal{C}: [x' y']\underbrace{\begin{bmatrix} 100 & 0 \\ 0 & 25 \end{bmatrix}}_{Q'}\begin{bmatrix} x' \\ y' \end{bmatrix} + [25 \ \ 50]\begin{bmatrix} x' \\ y' \end{bmatrix} + 25 = 0$$

$$\mathcal{C}: 100x'^2 + 25y'^2 + 25x' + 50y' + 25 = 0$$

$$\mathcal{C}: 4x'^2 + y'^2 + x' + 2y' + 1 = 0$$

$$\mathcal{C}: 4\left(x'^2 + \frac{x'}{4} + \frac{1}{64}\right) - \frac{1}{16}\left(y'^2 + 2y' + 1\right) - 1 + 1 = 0$$

$$\mathcal{C}: 4\left(x' + \frac{1}{8}\right)^2 + (y'+1)^2 - \frac{1}{16} = 0$$

→ translate by vector $\left(-\frac{1}{8}, -1\right)$ to get rid of x' and y'.

$$x' = x'' - \frac{1}{8}$$
$$y' = y'' - 1$$

New frame: $\mathcal{K}'' = (O'', B'')$, $O''\left(-\frac{1}{8}, -1\right)$

$$\mathcal{C}: 4\left(x'' \cancel{-\frac{1}{8}} \cancel{+\frac{1}{8}}\right)^2 + \left(y'' \cancel{-1} \cancel{+1}\right)^2 - \frac{1}{16} = 0$$

$$\mathcal{C}: 4x''^2 + y''^2 - \frac{1}{16} = 0$$

$$\mathcal{C}: \frac{x''^2}{\frac{1}{64}} + \frac{y''^2}{\frac{1}{16}} - 1 = 0 \Rightarrow \text{„Tall" ellipse}$$

$\boxed{\text{Canonical form} \Rightarrow a > b}$ We want $\frac{1}{16}$ under x''^2

$\Rightarrow$ Permute the coord. axes using $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in SO(2)$

$$\Rightarrow \mathcal{C}: \mathcal{E}_{\frac{1}{4}, \frac{1}{8}}: \frac{x'''^2}{\frac{1}{16}} + \frac{x'''^2}{\frac{1}{64}} - 1 = 0.$$

PERBOLA

$C: \underbrace{-94x^2+360xy+263y^2}-91x+221y+169=0.$

$Q = \text{symm. matrix} = \begin{bmatrix} -94 & 180 \\ 180 & 263 \end{bmatrix}$

$C: [x \ \ y]\begin{bmatrix} -94 & 180 \\ 180 & 263 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} + [-91 \ \ 221]\begin{bmatrix} x \\ y \end{bmatrix} + 169 = 0$

$\begin{vmatrix} -94-\lambda & 180 \\ 180 & 263-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 338 \Rightarrow v_1 = (5,12)$

$\Rightarrow \lambda_2 = -169 \Rightarrow v_2 = (12,-5)$

$\Rightarrow \begin{bmatrix} 5 & 12 \\ 12 & -5 \end{bmatrix} \rightarrow \det = -25 - 144 = -169$

$\|v_1\| = \sqrt{5^2+12^2} = \sqrt{169} = 13$

$\Rightarrow M_{BB'} = \frac{1}{13}\begin{bmatrix} 5 & 12 \\ 12 & -5 \end{bmatrix} \rightarrow \det = -1$

Make $\det = 1 \Rightarrow M_{BB'} = \frac{1}{13}\begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix} \in SO(2)$

$C: [x' \ \ y'] \, M_{BB'}^T \cdot Q \cdot M_{BB'} \begin{bmatrix} x' \\ y' \end{bmatrix} + b \cdot M_{BB'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + 169 = 0$

$C: [x' \ \ y'] \cdot \underbrace{\begin{bmatrix} 338 & 0 \\ 0 & -169 \end{bmatrix}}_{Q'} \begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{[169 \ \ 169]}_{b'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + 169 = 0$

$C: 338x'^2 - 169y'^2 + 169x' + 169y' + 169 = 0$

$C: 2x'^2 - y'^2 + x' + y' + 1 = 0$

$C: 2\left(x'^2 + \frac{x^2}{2} + \frac{1}{16}\right) - \frac{1}{8} - \left(y'^2 - y' + \frac{1}{4}\right) + \frac{1}{4} + 1 = 0$

$C: 2\left(x' + \frac{1}{4}\right)^2 - \left(y' - \frac{1}{2}\right)^2 + \frac{9}{8} = 0.$

$\rightarrow$ translate by vector $\left(-\frac{1}{4}, \frac{1}{2}\right) \Rightarrow K'' = (O'', B''), \ O'' = \left(-\frac{1}{4}, \frac{1}{2}\right)$

$x'' = x' - \frac{1}{4}$

$y'' = y' + \frac{1}{2}$

$C: 2 \cdot x''^2 - y''^2 + \frac{9}{8} = 0 \quad /: \frac{9}{8}$

$C: -\frac{x''^2}{\frac{9}{16}} + \frac{y''^2}{\frac{9}{8}} - 1 = 0$ $\Rightarrow$ „TALL Hyperbola"

$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in SO(2)$

$\Rightarrow C: \mathcal{H}\ldots : \frac{x'''^2}{\frac{9}{8}} - \frac{y'''^2}{\frac{9}{16}} - 1 = 0$

Quiz 5

~~1. projections are NOT invertible.~~
~~Homog. matrix in $\mathbb{E}^3$ is 4×4 - true $\boxed{B}$~~
~~Reflections are invertible - true $\boxed{C}$~~
~~Tensor prod is always sym - false.~~

~~2. -~~

$\boxed{\text{PARABOLA}}$ $\mathcal{C}: \underbrace{128x^2 + 480xy + 450y^2}+ 391x + 119y + 289 = 0.$

$$Q = \begin{bmatrix} 128 & 240 \\ 240 & 450 \end{bmatrix}$$

$$\mathcal{C}: [x\ y]\begin{bmatrix} 120 & 240 \\ 240 & 450 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} + [391\ \ 119]\begin{bmatrix} x \\ y \end{bmatrix} + 289 = 0$$

$$\begin{vmatrix} 120-\lambda & 240 \\ 240 & 450-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 578 \Rightarrow (8,15) = v_1$$
$$\Rightarrow \lambda_2 = 0 \Rightarrow (15,-8) = v_2$$

$$\|v_1\| = \sqrt{8^2+15^2} = 17$$

$$\Rightarrow \underset{\sim}{M_{BB'}} = \frac{1}{17}\begin{bmatrix} 8 & 15 \\ 15 & -8 \end{bmatrix} \cancel{\in SO(2)} \in O(2) \rightarrow \det = -1$$

$\rightarrow$ make it have $\det = 1$ $\rightarrow$ $M_{BB'} = \frac{1}{17}\begin{bmatrix} 8 & -15 \\ 15 & 8 \end{bmatrix} \in SO(2)$

$$\mathcal{C}: [x'\ y']\, M_{BB'}^T \cdot Q \cdot M_{BB'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + b \cdot M_{BB'} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + 289 = 0.$$

$$\mathcal{C}: [x'\ y'] \cdot \underbrace{\begin{bmatrix} 578 & 0 \\ 0 & 0 \end{bmatrix}}_{Q'}\begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{\begin{bmatrix} 289 & -289 \end{bmatrix}}_{b'}\begin{bmatrix} x' \\ y' \end{bmatrix} + 289 = 0.$$

$$\mathcal{C}: 578x'^2 + 289x' - 289y' + 289 = 0$$

$$\mathcal{C}: 2x'^2 + x' - y' + 1 = 0$$

$$\mathcal{C}: 2\left(x'^2 + \frac{x'}{2} + \frac{1}{16}\right) - \frac{1}{8} - y' + 1 = 0$$

$$\mathcal{C}: 2\left(x' + \frac{1}{4}\right)^2 - \left(y' - \frac{7}{8}\right) = 0$$

$\rightarrow$ translate by vector $v = \left(-\frac{1}{4}, \frac{7}{8}\right)$ $\rightarrow$ new frame: $K'' = (O'', B'')$

$x'' = x' - \frac{1}{4},\ y'' = y' + \frac{7}{8}$ $\qquad O'' = \left(-\frac{1}{4}, \frac{7}{8}\right)$

$\Rightarrow \mathcal{C}: 2x''^2 - y'' = 0 \iff x''^2 = \frac{1}{2}y''$; $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in SO(2)$ for permuting the coord. axes.

$\Rightarrow \mathcal{C}: \mathcal{P}_{\frac{1}{4}}: y'''^2 = 2 \cdot \frac{1}{4}x'''$